Fair Allocation of Job Training under Uncertain Treatment Benefits

ICT 5143: Algorithms Fairness and Graphical Models Laboratory
Problem 6 Team 6
Akshay Wali (261090830018) Mohit Das (261090840018)
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Abstract

We ask whether requiring evidence of nonnegative benefits for each recorded race group changes job-training allocations under a fixed budget. We use the randomized NSW experiment, a causal treatment–earnings DAG, and cross-fitted doubly robust policy scores. The lower-confidence-bound rule was infeasible in all five original folds at 40% capacity. At 20%, the point-estimate floor changed minimum group benefit by 13.22 earnings units (paired 95% interval $[-1231.76, 725.46]$). Sparse subgroup support limits conclusions about fairness and utility.

1 Question, context and contribution

Research question. At fixed capacity, can a lower-confidence-bound (LCB) group floor protect the group with the lowest incremental earnings benefit while preserving program value? Program administrators allocate scarce places, while job seekers bear the consequences of selection. A hypothetical offer is $D = 1$. The binary outcome used only in secondary fairness and calibration audits is $Y_+ = 1$ when 1978 real earnings are positive. Policy value uses continuous earnings.

Contribution. With the same candidate rules and budget, we test whether replacing a point-estimate floor with a simultaneous LCB reduces selections that violate a known group floor when simulated minority samples are small. An equal or higher violation frequency would refute the claim in that setting. We also count infeasible fits and zero-group allocations.

Closest work. The unconstrained baseline is a restricted NSW adaptation of Athey and Wager's doubly robust empirical welfare maximization [1]. We use a finite stump grid and compare selection with and without group-benefit constraints. Fair Policy Targeting provides a conceptual comparison for group welfare [3]. See `docs/related_work.md` for the eight-paper comparison.

2 Data, graphical model and fairness method

Data and estimand. The Dehejia–Wahba experimental RE74 subset has 445 historical male participants: 185 treated and 260 controls [2]. The analysis retains all rows and splits by participant. W is the observed pretreatment vector, including age and the recorded Black and Hispanic indicators. Race groups are Black, Hispanic, and neither indicator recorded. Age bands and race-by-age groups are secondary audits. T is historical training assignment and Y is observed RE78. The unobserved potential outcome is missing for each participant.

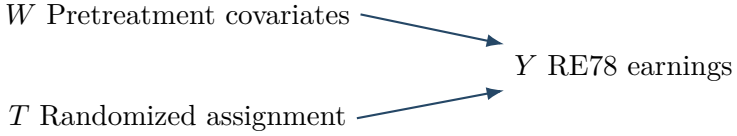


Figure 1. Experimental NSW causal DAG. The separate selection simulations are documented in `docs/causal_diagram.md`.

Graph and inference. Randomization, consistency, positivity and no interference support an empty back-door adjustment set. With d a hypothetical rule and G the recorded race group,

$$p(w, t, y) = p(w)p(t)p(y \mid w, t), \qquad B_g(d) = \mathbb{E}[d(W)\{Y(1) - Y(0)\} \mid G = g].$$

The project computes adjusted effects and cross-fitted doubly robust policy values. In a separate discrete inference fixture, variable elimination and clique-tree inference both return $P(Y = 1 \mid T = 1, X = 1) = 0.70$.

Method and fairness. The policy maximizes incremental earnings per eligible participant at fixed capacity. Primary fairness is $\min_g B_g(d)$ because aggregate gains can hide a group's low benefit. Allocation gaps and outcome-conditioned fairness are diagnostics. The main validation floor is zero and the acceptable utility loss is 200 earnings units per eligible person. Recorded race informs constrained validation and ridge predictions. The final stumps use no race indicator and use age only when selected as a split.

3 Evaluation protocol

The five treatment-stratified outer folds each contain 267 development, 89 validation, and 89 test participants. Candidate stumps are fitted using development data, selected with validation scores and evaluated on the held-out fold. Outcomes and input scaling used for fitting come from the corresponding training data. The unconstrained, point-estimate, and LCB rules share the candidate class and capacity. Other comparators include random and equal-race-rate allocation, predicted-earnings and estimated-benefit rankings, and a depth-two earnings tree against ridge on the same test folds.

We estimate the experimental ATE, adjusted outcome regression and IPW. We evaluate policies, simulate known heterogeneous effects and study overlap under simulated selection. Ablations replace the LCB with a point estimate, remove the floor, or rank by predicted earnings. Robustness checks vary capacity (20%, 40%, 60%), propensity clipping, earnings transformation and hidden confounding. Stochastic simulations and refits use seeds 11, 22, 33, 44 and 55. Available paired 95% intervals use 2,000 participant bootstrap replicates. Exact inference agreement used tolerance 10^{-8} and the largest discrepancy was 5.55×10^{-17}. The protocol is retrospective. Holm correction is reserved for a future independent confirmatory evaluation.

4 Results and interpretation

Rule at 40% capacity	Utility	Min. benefit	Feasible folds	Feasible refits
Unconstrained (closest baseline)	1,074.67	88.80	5/5	500/500
LCB floor (proposed)	-	-	0/5	0/500
Point-estimate floor (ablation)	-	-	2/5	158/500

Table note. Utility and minimum race-group benefit are raw earnings gains per eligible participant overall and within the worst-off group, respectively. Higher is better. The target test population is 445 participants pooled across five folds. Full pooled values require a policy in every fold. Brier and log-score diagnostics apply to the separate earnings model. The LCB rule selected no feasible policy. The point-estimate rule selected a policy in only two of the five folds.

Main finding. The LCB rule had no feasible selection. At 40% capacity, one point-estimate fold and zero LCB folds met the 200-unit validation tolerance. No full-sample proposed-minus-baseline held-out contrast was available. At 20%, where all point-estimate and unconstrained folds were evaluable, the point-minus-unconstrained differences were -155.37 for utility (95% interval $[-531.87, 161.54]$) and 13.22 for minimum group benefit (95% interval $[-1231.76, 725.46]$).

Predictive and probabilistic results. The experimental earnings difference was 1,794.34 (95% interval $[518.53, 3132.21]$). Ridge and the shallow earnings tree had held-out MAE/RMSE of 4912.74/6807.66 and 4880.91/6805.01. Both paired error-difference intervals included zero. The separate zero-plus-lognormal diagnostic had a Brier score of 0.2132 for positive earnings, mean predictive log score of -7.5701, and 96.85% coverage for nominal 90% intervals.

Mechanism and failure. At 40%, equal-race-rate allocation had a 1.94 percentage-point race access gap, but its minimum estimated group gain was −522.33. The Black, Hispanic, and neither-indicator groups contained 371, 39, and 35 people. The latter two had treated/control effective sizes of 11/28 and 18/17, all below 30. Treat those arm-level estimates and sparse intersections as exploratory.

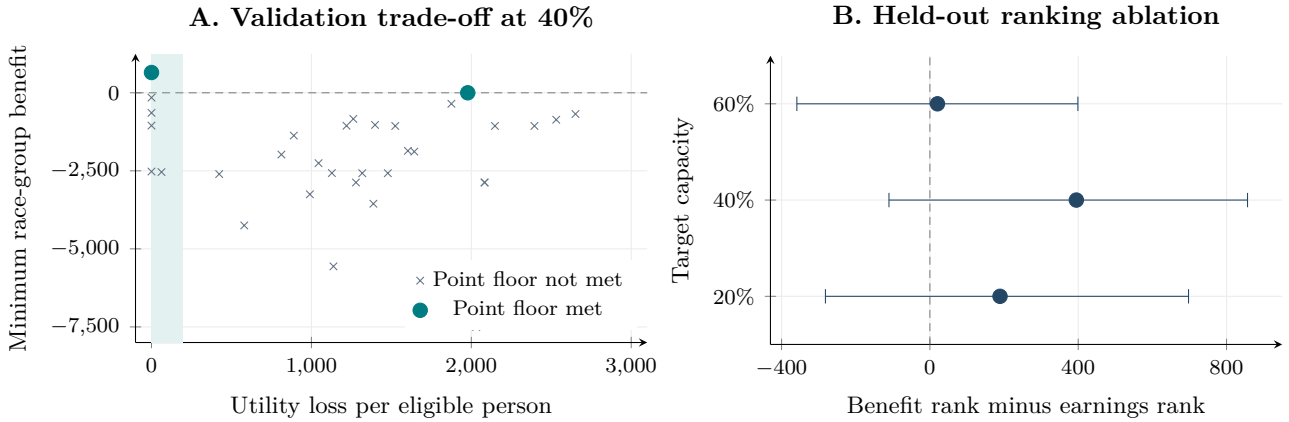


Figure 2. Panel A pools displayed candidates across five validation folds. Utility loss is relative to the maximum in each candidate's own fold. The shaded band is the 200-unit tolerance. No candidate passed the LCB floor. Panel B shows paired differences in raw earnings gains per eligible participant and pointwise 95% bootstrap intervals. The fold-specific frontier is in `results/figures/fairness_utility_frontier_40pct.pdf`.

5 Limitations, ethics and conclusion

The historical male sample limits transportability and does not support a gender fairness audit. Recorded race indicators are coarse, and prior earnings may reflect structural barriers. A zero benefit floor can coexist with zero access. In the separate zero-floor simulation with minority $n = 20$, the point rule produced 9 true violations in 500 datasets. The LCB rule selected in 432 datasets, with zero violations and zero places for at least one group in 94.21% of its selections. Thus the simulation shows how stricter selection can reduce observed violations while limiting access. The NSW data leave the benefit–utility claim unresolved. A next experiment would test an explicit minimum-access rule under the same budget.

Reproducibility and individual contributions

Reproduce. With verified NSW inputs and locked dependencies, run `.\run_all.ps1` in Windows PowerShell with Python 3.14.6. Configuration is in `configs/frozen_run.json`. The pinned rerun verified 40 artifact hashes and seven exact CSV matches.

Akshay Wali. Set up the data and environment, ran the experiments, and verified the clean reproduction. **Mohit Das.** Individual work and peer cross-check are pending confirmation. **Evidence.** `docs/requirements_matrix.md` and `results/manifest.csv`.

References

[1] S. Athey and S. Wager. Policy learning with observational data. *Econometrica*, 89(1), 133–161, 2021. DOI 10.3982/ECTA15732.

[2] R. Dehejia and S. Wahba. Causal effects in nonexperimental studies. *JASA*, 94(448), 1053–1062, 1999. NSW RE74 dataset documentation, checked 8 November 2026.

[3] D. Viviano and J. Bradic. Fair policy targeting. *JASA*, 119(545), 730–743, 2024. DOI 10.1080/01621459.2022.2142591.